

2023 Ph H2 Q9

Section: Particles and Waves

Topic: Photoelectric Effect — Analogy

Summary of Question:

Evaluate a vending-machine analogy for the photoelectric effect: a snack only appears if enough money is inserted; extra money gives change.

Comment using physics

- Good match (threshold): There is a minimum energy per photon required to liberate an electron: $E = hf \geq \phi$. This is like the machine price — if a single photon has too little energy (f below threshold), no electrons are emitted regardless of how many such photons arrive.
- Good match (excess energy $\rightarrow$ KE): Any excess energy per photon becomes electron kinetic energy: $K_{\text{max}} = hf - \phi$. This is analogous to ‘change’ from paying more than the price.
- Limits of the analogy (intensity): Increasing the number of below-threshold photons (more coins of too small a value) still does not eject electrons — correct. Above threshold, increasing intensity at fixed frequency gives more photoelectrons per second (more ‘snacks’ dispensed), but does not increase their maximum KE; in a vending machine, paying more than once usually gives multiple snacks rather than identical ‘change’.

- Other limitations: The analogy cannot capture the lack of time delay (emission is immediate at threshold) and it might suggest multi-coin addition to reach the price, whereas in the photoelectric effect a single photon interacts with a single electron under normal conditions.

Final Answer

The analogy is partly valid: the threshold requirement and the idea that excess goes into KE ('change') are good. However, it is limited because intensity affects the number of electrons emitted (not their KE at fixed f), and photoemission is a one-photon-one-electron process with no time delay.

Revision Tips

- Key equations: $E = hf$; $K_{\max} = hf - \phi$; $eV_{\text{stop}} = K_{\max}$.
- Frequency sets KE; intensity sets photoelectron rate (if $f > \text{threshold}$).
- Below threshold frequency: no emission regardless of intensity.